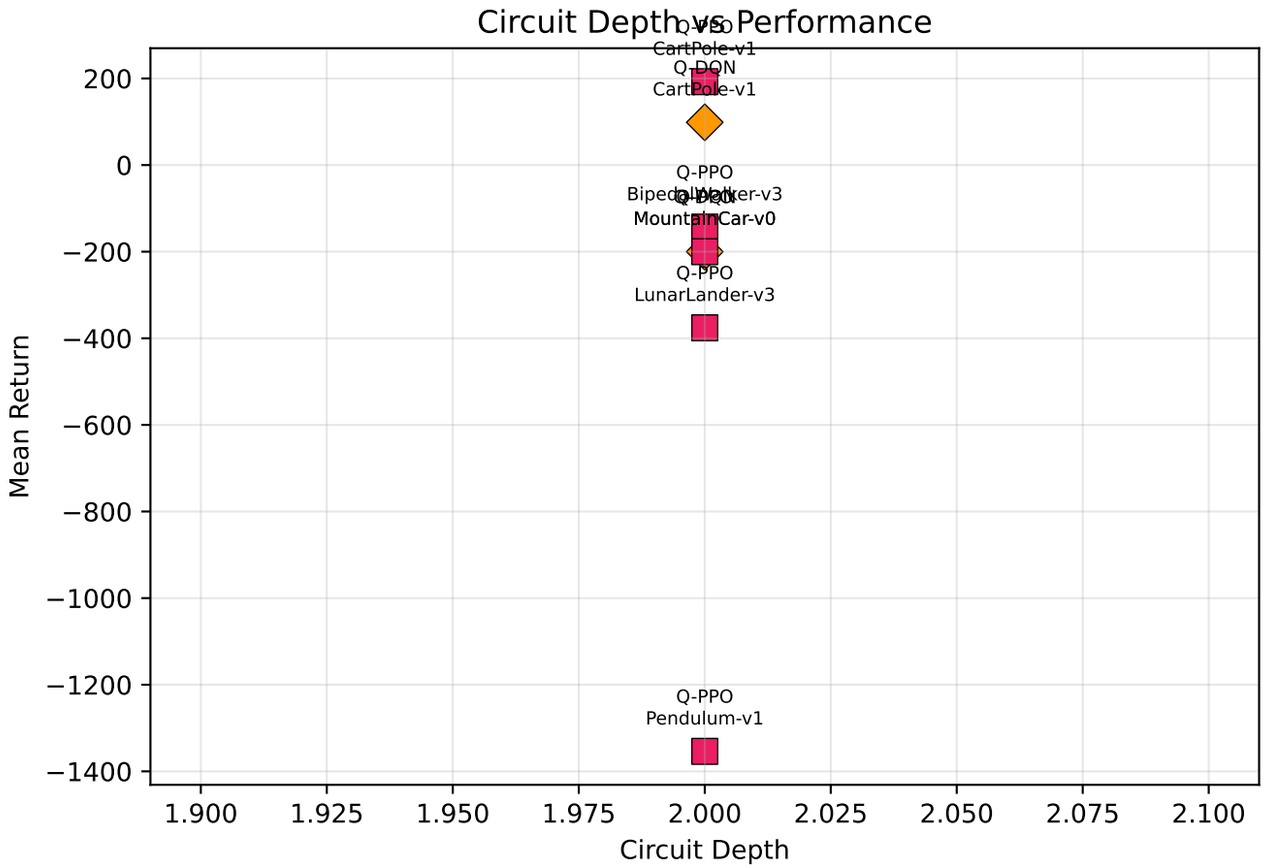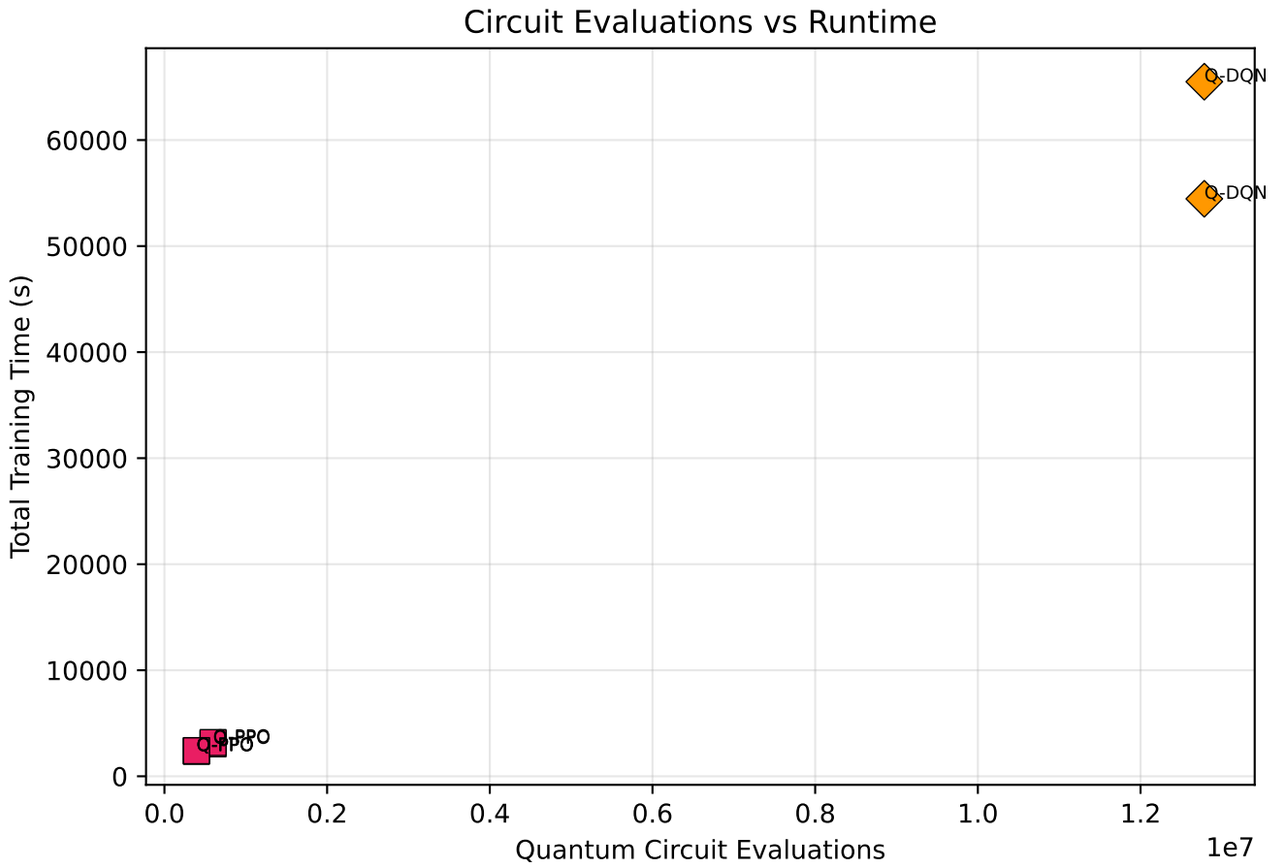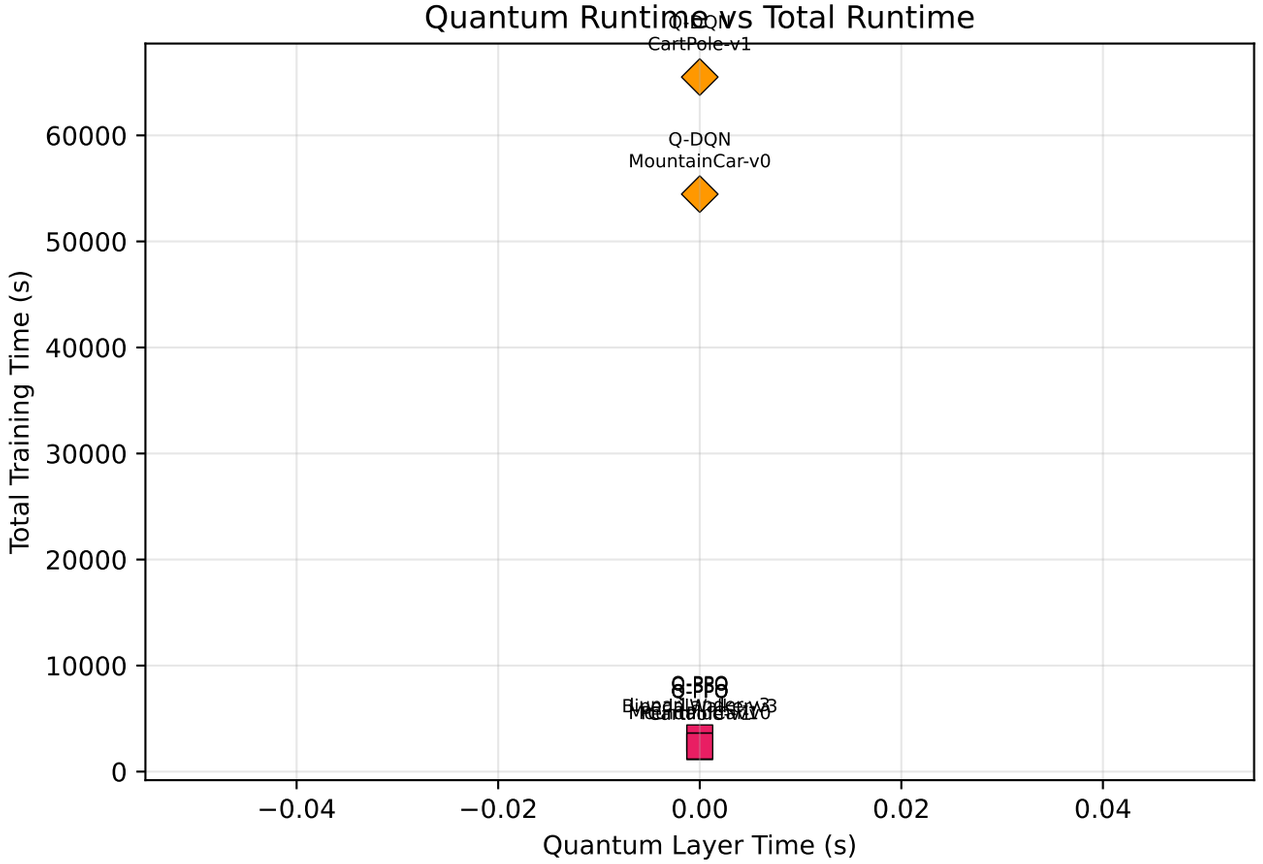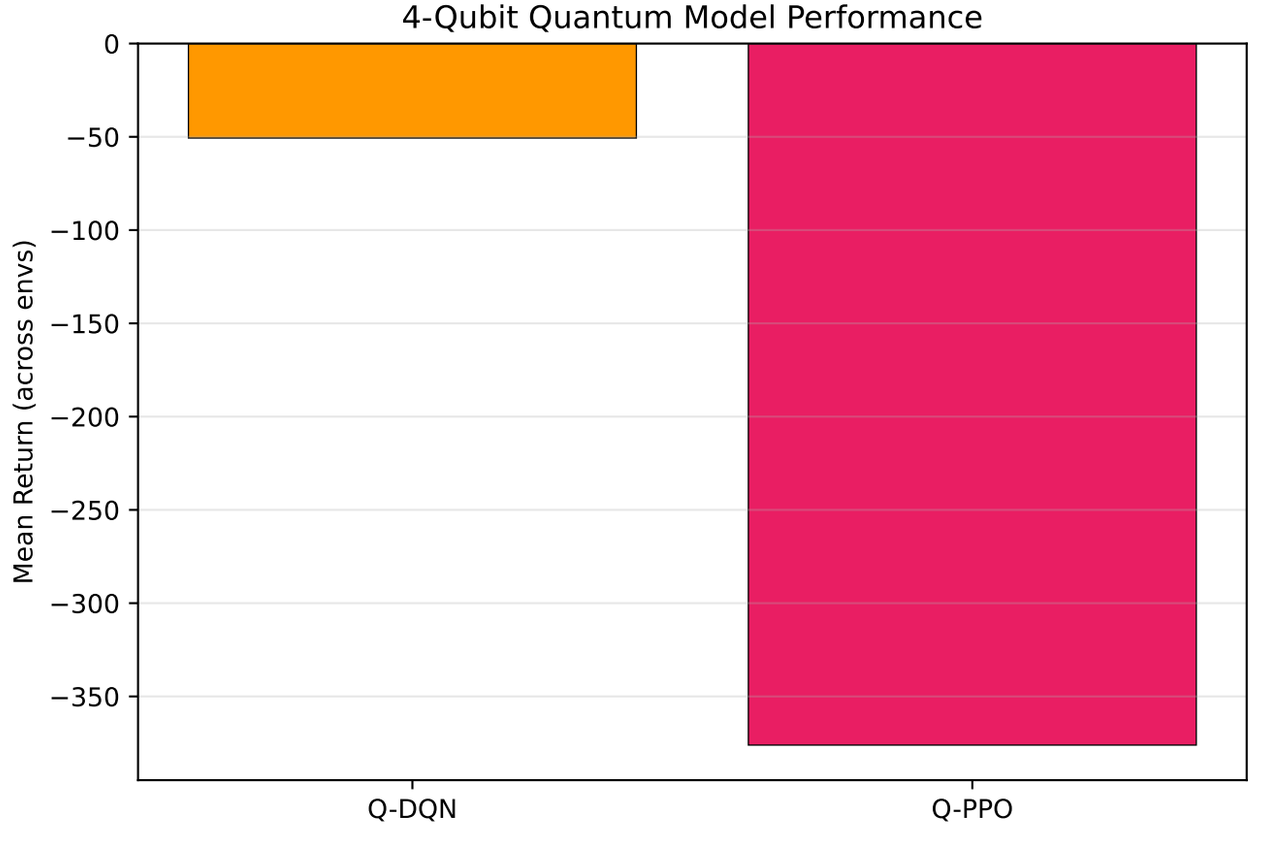Quantum Resource Analysis